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An enhanced sensitivity analysis framework for residual ionospheric error in network RTK geometry-free three-carrier ambiguity resolution

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Abstract

With the rapid development of multi-frequency Global Navigation Satellite Systems (GNSS), Geometry-Free (GF) Three-Carrier Ambiguity Resolution (TCAR) has become an increasingly viable solution for high-precision positioning. The GF TCAR model enables satellite-by-satellite ambiguity resolution, offering robustness independent of satellite geometry. However, its performance in network Real-Time Kinematic (RTK) applications is highly sensitive to residual atmospheric errors. To accurately quantify these impacts and optimize ambiguity resolution strategies, this study proposes an enhanced analysis framework and presents a comprehensive sensitivity analysis of the GF TCAR model under different atmospheric and noise conditions. Theoretical derivations and simulations confirm that non-dispersive errors, such as tropospheric delays, are effectively mitigated within the GF framework. In addition, the study establishes quantitative accuracy thresholds for ionospheric modeling across constellations. While Extra-Wide-Lane (EWL) and Wide-Lane (WL) ambiguities remain robust against ionospheric errors of up to dozens of Total Electron Content Units (TECU) and several TECU, respectively, Narrow-Lane (NL) ambiguity resolution is the most challenging step, requiring residual errors to remain about within ± 0.2 TECU. Furthermore, the analysis shows that EWL ambiguity resolution is relatively insensitive to measurement noise, whereas WL ambiguity resolution is predominantly noise-limited, especially as the EWL wavelength increases. These findings provide valuable error-tolerance references for the quality control of ionospheric corrections in network RTK services.

Keywords GNSS, Geometry-free model, TCAR, Network RTK, Ionospheric delay, Sensitivity analysis

Introduction

High-precision positioning has become essential across a wide range of applications, from geodetic surveying to autonomous vehicle navigation (Chen et al., 2024). Global Navigation Satellite Systems (GNSS) based algorithms,

such as Real-Time Kinematic (RTK), are fundamental to achieving centimeter-level precision (Kaplan and Hegarty 2017; Odolinski et al., 2015). Correctly resolving the integer ambiguities is a prerequisite for these carrier-phase based positioning algorithms (Geng et al., 2009; Teunissen et al., 1999).

To facilitate reliable ambiguity resolution, various methods have been developed, which are generally categorized into Geometry-Based (GB) and Geometry-Free (GF) models (Teunissen et al., 2002). In the GB model, ambiguities are estimated jointly with baseline components (and atmospheric errors), using the integer least squared estimation (Li, 2018). This approach utilizes

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the geometric relationships between the receiver and multiple tracked satellites, thereby obtaining a higher ambiguity resolution success rate than that with the GF model (Teunissen et al., 2002). In contrast, the GF model resolves the ambiguity for each satellite pair independently, which reduces computational complexity and ensures the independence of ambiguity correctness for each pair (Cheng et al., 2024). Therefore, improving the ambiguity resolution success rate in the GF model is of significant importance.

With the advent of modern GNSS constellations, such as Global Positioning System (GPS), Galileo, and BeiDou satellite System (BDS), which now broadcast three- or even five-frequency signals (Zhang, 2021), the implementation of GF models has become increasingly feasible. Two representative GF methods are Three-Carrier Ambiguity Resolution (TCAR) (Forssell et al., 1997; Vollath et al., 1999) and Cascaded Integer Resolution (CIR) (Hatch et al., 2000; Jung et al., 2000). Both methods are based on the virtual carrier phases generated through linear combinations of different frequencies, resulting in longer virtual wavelengths and smaller noise-to-wavelength ratios. (Han & Rizos, 1999). These virtual wavelengths are classified as Extra-Wide-Lane (EWL: $\lambda \geq 2.93$ m), Wide-Lane (WL: $0.75 \text{ m} < \lambda < 2.93$ m), and Narrow-Lane (NL: $\lambda \leq 0.75$ m) (Xu & Chen, 2018). Due to the long wavelength of EWL, its ambiguity can be directly resolved using pseudorange measurements. The resolved EWL ambiguities can then be used to fix WL and NL ambiguities (Feng & Li, 2008).

Extensive research has been conducted on this topic since the advent of multi-frequency GNSS signals. Ji et al. (2007), Feng (2008), Cocard et al. (2008), and Li et al. (2017) have investigated optimal frequency combinations for different constellations, thereby enhancing the performance of TCAR. These studies have demonstrated that instantaneous ambiguity resolution is highly probable for short baselines (Zhang & He, 2016), which is important for real-time applications such as navigation. However, the applicability of short-baseline TCAR, typically limited to baselines shorter than 10 km, remains constrained. For medium and long baselines, the primary challenge in resolving NL ambiguities lies in atmospheric delays.

To address, this issue, several models have been proposed, including modified TCAR (Tang et al., 2014; Tian et al., 2018; Zhao et al., 2015), the Geometry-Ionosphere-Free (GIF) model (Li et al., 2010), and atmospheric estimation models (Wang et al., 2016, 2020). Although these models differ in formulation, they all use the combinations of pseudorange, fixed EWL, and WL ambiguities to estimate atmospheric delays for NL ambiguity resolution. However, the accuracy of these

estimations is often insufficient when derived from single-epoch data. Consequently, multi-epoch measurements are typically employed to smooth the estimated atmospheric delays. Once the smoothed atmospheric corrections reach sufficient accuracy, the corresponding NL ambiguities can be reliably resolved. While most studies evaluated these models based on the required time for smoothing, few have focused on the accuracy requirements for the smoothed atmospheric estimations.

In addition to these smoothing techniques, network RTK has also been developed to correct atmospheric delays at a rover station (Berber et al. 2013; Li et al., 2014). In the network RTK, at least three reference stations are required to interpolate atmospheric delays (Hong et al., 2022). Once the atmospheric corrections derived from the network RTK reference stations are applied, this approach always assumes that residual atmospheric errors are negligible. In this case, ambiguity resolution at the rover can be performed in a manner similar to that for a short baseline (Pehlivan et al., 2019). Much research has been devoted to improving the accuracy of interpolated atmospheric corrections (Al-Shaery et al., 2011; Cui et al., 2018; Ouassou et al., 2015; Tang et al., 2016). However, the presence of tropospheric and ionospheric gradients renders the interpolated corrections less accurate than anticipated. (Li et al., 2025; Xu et al., 2024). This effect is particularly significant in low-latitude regions, such as Hong Kong, where ionosphere activity is especially pronounced. Therefore, investigating the impact of these residual errors and noise on the performance of the GF TCAR model is a topic worthy of further research in network RTK.

In this paper, we propose an enhanced sensitivity analysis framework and systematically investigate the accuracy requirements for residual tropospheric and ionospheric errors to ensure reliable ambiguity resolution using the GF TCAR method. We first utilize a representative GF TCAR model to demonstrate the analytical process, followed by a theoretical derivation to estimate the necessary accuracies of atmospheric corrections. These theoretical findings are then validated through a series of simulation experiments and real-world network RTK data. It should be emphasized that the GF TCAR model employed here serves as an example to demonstrate the proposed analysis framework. The established sensitivity analysis framework is transferable and can be adapted to assess the robustness of other GF model variants or linear combinations. Consequently, the benefits of this study provide not only practical atmospheric correction accuracy requirements but also a standardized framework for optimizing GF model ambiguity resolution across modern multi-frequency GNSS.

TCAR model

When applying the GF TCAR method to medium or long baselines, the primary challenge lies in mitigating tropospheric and ionospheric delays. In this section, it is assumed that atmospheric delays are corrected using multi-station interpolation within the network RTK framework. As a result, the intra-constellation double-difference measurements employed in the subsequent processing are considered to be largely free from atmospheric effects. However, it should be noted that, in practice, some residual atmospheric errors may still exist. The corrected double-difference equations are:

$$P_i = \rho + \delta T + \delta I_i + \varepsilon_i \quad (1)$$

$$\phi_i = \rho + \delta T - \delta I_i + \lambda_i N_i + \epsilon_i \quad (2)$$

where, the double-difference operator $\Delta\nabla$ and indexes of satellite and receiver are omitted for brevity. The subscript i denotes the frequency f_i . P and ϕ represent the double-difference pseudorange and carrier phase observations, respectively, both expressed in meters. ρ denotes the double-difference geometric distance. δT and δI represent the residual slant tropospheric and ionospheric errors, respectively, remaining after applying the network RTK corrections. The double-difference satellite orbit error is typically negligible and can be absorbed into δT , and is therefore ignored. λ is the carrier wavelength corresponding to frequency f_i , and N is the double-difference integer ambiguity. ε and ϵ denote code and carrier measurement noises. It should be noted that the ionospheric error, denoted as δI , is frequency dependent. Given that the first-order term accounts for more than 99.9% of the total ionospheric effect, it can be expressed in meters as (Timote et al., 2024):

$$\delta I_i = \frac{40.3 \times 10^{16}}{f_i^2} \times E_{\text{TEC}} \quad (3)$$

where E_{TEC} represents the residual slant Total Electron Content (TEC). It is expressed in TEC Units (TECU), where 1 TECU is equivalent to 1×10^{16} electrons/m².

For clarity, the three selected frequencies are denoted as f_1 , f_2 and f_3 . Taking GPS as an example, f_1 , f_2 and f_3 correspond to the frequencies of L1 (1575.42 MHz), L2 (1227.60 MHz), and L5 (1176.45 MHz), respectively. The linear combination of three fundamental signals can be generally formulated as (Feng & Rizos, 2005):

$$\phi_{(i,j,k)} = \frac{k \bullet f_1 \bullet \phi_1 + l \bullet f_2 \bullet \phi_2 + m \bullet f_3 \bullet \phi_3}{k \bullet f_1 + l \bullet f_2 + m \bullet f_3} \quad (4)$$

where, k , l and m are integer coefficients. The corresponding virtual frequency $f_{(k,l,m)}$ and wavelength $\lambda_{(k,l,m)}$ of linearly combined signal are (Feng, 2008):

$$f_{(k,l,m)} = k \bullet f_1 + l \bullet f_2 + m \bullet f_3 \quad (5)$$

$$\lambda_{(k,l,m)} = \frac{c}{f_{(k,l,m)}} \quad (6)$$

where c is the speed of light in vacuum.

Although there are different frameworks of the GF TCAR steps, we only analyze a typical model here. Other models can be evaluated in the same way.

Due to the relatively long wavelength of the EWL, its ambiguity can be directly fixed using pseudorange measurements, and the residual atmospheric errors are ignored at this step:

$$N_{(0,1,-1)} = f_{\text{round}} \left(\frac{\phi_{(0,1,-1)} - P}{\lambda_{(0,1,-1)}} \right) \quad (7)$$

with

$$\lambda_{(0,1,-1)} = \frac{c}{f_2 - f_3} \quad (8)$$

$$\phi_{(0,1,-1)} = \left(\frac{\phi_2}{\lambda_2} - \frac{\phi_3}{\lambda_3} \right) \times \lambda_{(0,1,-1)} \quad (9)$$

$$P = \frac{P_1 + P_2 + P_3}{3} \quad (10)$$

where, $f_{\text{round}}()$ refers to a rounding function. Subsequently, the WL ambiguity can be directly fixed using the previously resolved EWL ambiguity:

$$N_{(1,-1,0)} = f_{\text{round}} \left(\frac{\phi_{(1,-1,0)} - P_{(0,1,-1)}}{\lambda_{(1,-1,0)}} \right) \quad (11)$$

with

$$\lambda_{(1,-1,0)} = \frac{c}{f_1 - f_2} \quad (12)$$

$$\phi_{(1,-1,0)} = \left(\frac{\phi_1}{\lambda_1} - \frac{\phi_2}{\lambda_2} \right) \times \lambda_{(1,-1,0)} \quad (13)$$

$$P_{(0,1,-1)} = \phi_{(0,1,-1)} - \lambda_{(0,1,-1)} N_{(0,1,-1)} \quad (14)$$

To mitigate the impact of noise, all three-frequency carrier phase measurements are used for ambiguity resolution in the final step. With the ambiguities $N_{(1,-1,0)}$ and $N_{(0,1,-1)}$ already being fixed, we have:

$$\begin{cases} N_{(1,-1,0)} = N_1 - N_2 \\ N_{(0,1,-1)} = N_2 - N_3 \\ \phi_1 = \rho + \delta T - \delta I_1 + \lambda_1 N_1 + \epsilon_1 \\ \phi_2 = \rho + \delta T - \delta I_2 + \lambda_2 N_2 + \epsilon_2 \\ \phi_3 = \rho + \delta T - \delta I_3 + \lambda_3 N_3 + \epsilon_3 \end{cases} \quad (15)$$

Ignoring the δT and δI , the final solution ($\tilde{\rho}\tilde{N}_1\tilde{N}_2\tilde{N}_3$) can be obtained with the integer least squared method. The $\tilde{\rho}$ is the estimated double-difference geometric distance.

Theoretical analysis of the impact of different errors on TCAR

Although the previous section noted that residual atmospheric errors in network RTK can often be neglected, thereby simplifying the design of ambiguity resolution algorithms, these errors are not completely eliminated, even after applying network based corrections. This is particularly true for satellites at low elevation angles. Therefore, this section focuses on analyzing the impact of residual tropospheric and ionospheric errors, as well as measurement noise, on the success rate of TCAR. Although it is widely acknowledged that tropospheric errors are completely eliminated in the GF model, we still provide a detailed analysis of this phenomenon for two primary reasons. First, such an analysis can help readers who are unfamiliar with the GF model better understand this property. Second, this analysis establishes a generalized analytical framework for evaluating tropospheric impacts when different combination coefficients are adopted in various GNSS linear combinations.

An enhanced analytical framework is proposed in this section. Within this framework, each step of ambiguity resolution must consider both the current linear combination and the preceding combination that facilitates the fixing process. This specific consideration is often ignored in many existing studies, which primarily focus on selecting combinations based on their individual noise levels. For example, when a fixed EWL ambiguity is utilized to facilitate WL resolution, the analysis should focus on the composite error of the integrated EWL/WL combination, rather than treating the EWL and WL errors separately. This is because the EWL and WL observables may be correlated.

Unlike traditional variance analysis, we introduces a new evaluation criterion for network atmospheric corrections from the perspective of the maximum tolerable range. This is because the variance based analysis method assumes a zero-mean distribution, which may not be true for ionospheric corrections in network RTK over short periods such as within one hour. These ionospheric corrections often exhibit strong temporal correlations. For instance, smoothed ionospheric corrections from the network may appear stable and exhibit a small variance,

yet still contain systematic biases. In this case, the success rate may be significantly overestimated when evaluated solely through variance-based analysis.

Following the steps defined earlier, the EWL ambiguity, which is characterized by a relatively long wavelength and low noise sensitivity, can be directly resolved using code measurements. Based on Eq. (7), the error (in cycles) of the EWL ambiguity induced by residual atmospheric errors and measurement noise can be expressed as follows:

$$\begin{aligned} & \frac{40.3 \times 10^{16}}{3c} \left(\frac{4f_3 - f_2}{f_3^2} + \frac{f_3 - 4f_2}{f_2^2} + \frac{f_3 - f_2}{f_1^2} \right) \times E_{\text{TEC}} \\ & - \frac{\epsilon_1 + \epsilon_2 + \epsilon_3}{3 \times \lambda_{(0,1,-1)}} + \frac{\epsilon_2}{\lambda_2} - \frac{\epsilon_3}{\lambda_3} \end{aligned} \quad (16)$$

One can see that residual tropospheric errors are effectively mitigated, indicating that tropospheric delay does not impact the ambiguity resolution of EWL in theory. However, residual ionospheric errors, as well as code and carrier phase measurement noises, do influence EWL ambiguity resolution. According to Eq. (16), the ionospheric coefficient is approximately 0.0066 cycles per TECU for GPS, showed in Table 1. Thus, provided that code and carrier phase measurement noises are negligible, the EWL ambiguity can be correctly resolved as long as the residual ionospheric error does not exceed 75.66 TECU (i.e., 0.5 cycles divided by 0.0066 cycles/TECU). This requirement is easy met in network RTK applications. Furthermore, in Eq. (16), code measurement noise is averaged and carrier phase noise is not amplified, ensuring that EWL ambiguities can be reliably fixed with a high success rate instantaneously, even in challenging environments.

Once the EWL ambiguity has been fixed, the more precise range derived from EWL combination can be utilized to resolve the WL ambiguity directly, thereby avoiding the impact of code errors on this step. According to Eq. (11), the corresponding error (in cycles) of the WL ambiguity is calculated as:

$$\begin{aligned} & -\frac{40.3 \times 10^{16}}{c} \left(\frac{1}{f_1} - \frac{1}{f_2} - \frac{1}{f_3} + \frac{f_1}{f_2 f_3} \right) \times E_{\text{TEC}} + \frac{\epsilon_1}{\lambda_1} \\ & - \frac{f_1 - f_3}{f_2 - f_3} \frac{\epsilon_2}{\lambda_2} + \frac{f_1 - f_2}{f_2 - f_3} \frac{\epsilon_3}{\lambda_3} \end{aligned} \quad (17)$$

Similarly, residual tropospheric error has a negligible effect on ambiguity resolution at this step. Only residual ionospheric error and carrier phase measurement noise directly impact the fixing performance, with the corresponding coefficients listed in Table 1. If the residual ionospheric error does not exceed 6.1 TECU (i.e., 0.5 cycles divided by

0.0820 cycles/TECU), the WL ambiguity can be correctly fixed provided that carrier phase noise is negligible. This theoretical threshold is easily achieved with network based corrections. However, it should be noted that carrier phase noise is amplified several times at this step, which can reduce the success rate of WL ambiguity resolution. According to Eq. (17), as the value of $f_2 - f_3$ decreases, the amplification factor for carrier noise increases. A smaller frequency separation between f_2 and f_3 corresponds to a longer EWL wavelength. Consequently, a longer EWL wavelength makes the WL ambiguity more susceptible to carrier phase noise.

Once the EWL and WL ambiguities have been correctly resolved, it becomes possible to fix the original ambiguities ($N_1 N_2 N_3$). For clarity, Eq. (15) is reformulated as follows:

$$\begin{cases} \phi_1 = \rho + \delta T - \delta I_1 + \lambda_1 (N_2 + N_{(1,-1,0)}) + \epsilon_1 \\ \phi_2 = \rho + \delta T - \delta I_2 + \lambda_2 N_2 + \epsilon_2 \\ \phi_3 = \rho + \delta T - \delta I_3 + \lambda_3 (N_2 - N_{(0,1,-1)}) + \epsilon_3 \end{cases} \quad (18)$$

The original ambiguities can be solved using the integer least squares method as follows:

$$\begin{cases} N_2 = f_{\text{round}} \left(\frac{3 \sum_{i=1}^3 (\lambda_i \phi_i) - (\sum_{i=1}^3 \lambda_i) (\sum_{i=1}^3 \phi_i)}{3 \sum_{i=1}^3 \lambda_i^2 - (\sum_{i=1}^3 \lambda_i)^2} \right) \\ N_1 = N_2 + N_{(1,-1,0)} \\ N_3 = N_2 - N_{(0,1,-1)} \end{cases} \quad (19)$$

Regarding the NL ambiguities, it is enough to evaluate only the success rate for N_2 , as the relationships between the three ambiguities are already constrained by the fixed EWL and WL ambiguities. The impact of measurement errors on N_2 (in cycles) can be calculated as follows:

$$N_{2_error} = \frac{3 \sum_{i=1}^3 (\lambda_i (\delta T - \delta I_i + \epsilon_i)) - (\sum_{i=1}^3 \lambda_i) (\sum_{i=1}^3 (\delta T - \delta I_i + \epsilon_i))}{3 \sum_{i=1}^3 \lambda_i^2 - (\sum_{i=1}^3 \lambda_i)^2} \quad (20)$$

When the value of Eq. (20) remains within ± 0.5 cycles, the original ambiguity can be correctly fixed. It can be calculated that the coefficient for the tropospheric error is zero, showed in Table 1, indicating that tropospheric delay does not affect the resolution of the original ambiguity. The coefficient for the ionospheric error is approximately 1.9808 cycles/TECU. Therefore,

if no noise, the original ambiguity can be correctly fixed as long as the residual ionospheric error does not exceed ± 0.252 TECU (i.e., 0.5 cycles divided by 1.9808 cycles/TECU). At this step, the coefficient associated with carrier phase noise is smaller than that for the WL ambiguity, indicating that, when the residual ionospheric delay is close to zero, the WL ambiguity may be the most difficult to resolve. However, as the residual ionospheric error increases, the original ambiguity becomes more difficult to resolve due to its higher sensitivity to ionospheric delay. In contrast, the EWL and WL ambiguities are less affected by residual ionospheric errors.

Success rate analysis of TCAR under different error sources

Based on the results presented above, this section evaluates the impact of various error sources on the TCAR ambiguity resolution success rate, using GPS as an example. To isolate the influence of each individual error, other errors are set to zero when analyzing a specific type of error. For theoretical simplicity, we assume that the standard deviations of code and carrier phase noise are the same for all three frequencies. While this assumption may not fully capture real-world noise characteristics, this limitation is addressed in the semi-simulation section, where atmospheric error analysis is conducted under real-world noise conditions using short-baseline GNSS data. The following mathematical expressions are defined to calculate the ambiguity resolution success rates:

$$\begin{cases} p_{EWL} = \int_{-0.5}^{0.5} \frac{1}{\sqrt{2\pi}\sigma_{EWL}} e^{-\frac{x^2}{2\sigma_{EWL}^2}} dx \\ p_{WL} = p_{EWL} \times \int_{-0.5}^{0.5} \frac{1}{\sqrt{2\pi}\sigma_{WL}} e^{-\frac{x^2}{2\sigma_{WL}^2}} dx \\ p_{NL} = p_{EWL} \times p_{WL} \times \int_{-0.5}^{0.5} \frac{1}{\sqrt{2\pi}\sigma_{NL}} e^{-\frac{x^2}{2\sigma_{NL}^2}} dx \end{cases} \quad (21)$$

Table 1 Coefficients of different error sources affecting GPS ambiguity resolution

Coefficients		of δT in cycle/m	of E_{TEC} in cycle/TECU	of ϵ_1 in cycle/m	of ϵ_2 in cycle/m	of ϵ_3 in cycle/m	of ϵ_1 in cycle/m	of ϵ_2 in cycle/m	of ϵ_3 in cycle/m
GPS	EWL	0	0.0066	-0.0569	-0.0569	-0.0569	0	4.10	-3.92
	WL	0	-0.0820	0	0	0	5.26	-31.94	26.69
	NL	0	1.9808	0	0	0	-16.49	6.03	10.46

where, p_{EWL} , p_{WL} and p_{NL} are success rates of ambiguity resolution; σ denotes the standard deviation.

It should be noted that code noise does not directly influence the resolution of the WL and original ambiguities in the defined TCAR framework. Therefore, this analysis focuses solely on evaluating the impact of code noise on the success rate of EWL ambiguity resolution. This simulated standard deviation of pseudorange ranges from 0 to 5 m. Large values are included here to represent degraded signal conditions in challenging environments, such as urban canyons. As illustrated in Fig. 1, when the standard deviation of code noise is below 1 m, the EWL ambiguity resolution success rate approaches 100%. However, as the code noise increases, the success rate gradually decreases.

Next, the influence of carrier phase noise on the success rates of different ambiguity resolutions is evaluated. As shown in Fig. 1, when the standard deviation of carrier phase noise increases from 0 to 30 mm, the EWL ambiguity resolution success rate remains nearly 100%. This demonstrates that EWL ambiguities are relatively robust and can be reliably fixed even in the presence of significant carrier phase noise. In contrast, the WL and NL ambiguities are more sensitive to carrier phase noise. Their success rates remain close to 100% only when the standard deviation is below 5 mm. When the standard deviation increases to 10 mm, the success rates for both WL and NL ambiguities decreases to about 70%. These results indicate that, in GNSS challenging environments,

it is impossible to correctly fix all ambiguities of WL and NL instantaneously.

Finally, the effect of ionospheric error on the success rates of different ambiguity resolutions is evaluated. As shown in Fig. 1, when the standard deviation of ionospheric errors increases from 0 to 3 TECU, the EWL ambiguity resolution success rate remains 100%. This finding is consistent with our previous analysis, confirming that EWL ambiguities are insensitive to ionospheric errors. For the WL ambiguity, the success rate remains almost 100% until the standard deviation exceeds 2 TECU, after which it gradually declines. In contrast, the NL ambiguity achieves a high success rate close to 100%, only when the standard deviation is below approximately 0.1 TECU. Beyond this threshold, the success rate for NL ambiguity resolution decreases rapidly. These results indicate that the original ambiguity is highly sensitive to ionospheric errors, emphasizing the necessity of properly accounting for ionospheric effects, especially for longer baselines.

Semi-simulation under different residual atmospheric error sources

In this section, to evaluate the impact of residual tropospheric and ionospheric errors on the performance of the GF TCAR model, a series of semi-simulated experiments are conducted. These simulated errors are added to GNSS data collected in a short baseline (9 m), where residual atmospheric effects can be considered negligible but

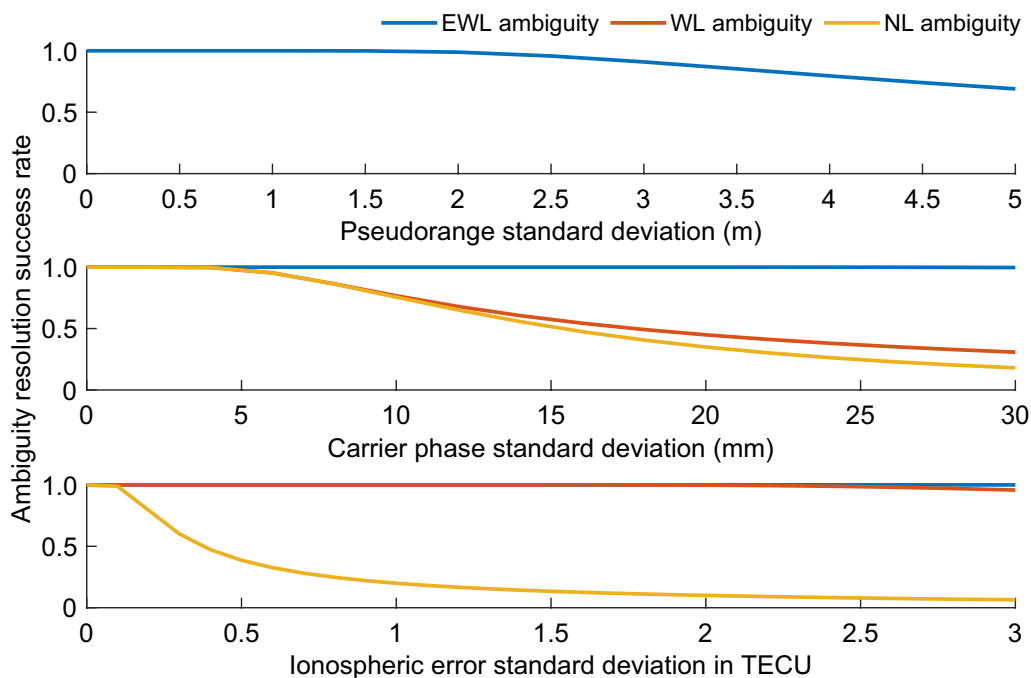


Fig. 1 GPS ambiguity resolution success rate of EWL, WL and NL under different error sources

Table 2 The details of the experimental setting

Base station	Antenna: Zhongyu HZZACF-S808; Receiver: Trimble BD990		
Rover station	Antenna: Zhongyu HZZACF-S808; Receiver: Trimble BD990		
Baseline	9 m		
Sampling duration	1 h		
Sample frequency	1 Hz		
Constellation	GPS	Galileo	BDS (II and III)
Carrier frequency	f_1 : 1575.42 MHz f_2 : 1227.60 MHz f_3 : 1176.45 MHz	f_1 : 1575.42 MHz f_2 : 1278.75 MHz f_3 : 1207.14 MHz	f_1 : 1561.098 MHz f_2 : 1268.52 MHz f_3 : 1207.14 MHz
EWL	$\lambda_{(0,1,-1)}$: 5.8610 m	$\lambda_{(0,1,-1)}$: 4.1865 m	$\lambda_{(0,1,-1)}$: 4.8842 m
WL	$\lambda_{(1,-1,0)}$: 0.8619 m	$\lambda_{(1,-1,0)}$: 1.0105 m	$\lambda_{(1,-1,0)}$: 1.0247 m

with real-world noise. Details of the GNSS data acquisition are provided in Table 2, with selected three frequencies from GPS, Galileo and BDS. Various simulated error sources are individually introduced to this GNSS dataset. To facilitate a clear analysis of each error type, the errors are added separately. It should be noted that simulated residual tropospheric and ionospheric errors are applied to the double-difference code and phase measurements for each satellite pair, respectively. Although the analysis is based on one-hour of data, the dataset comprises over 72,000 samples, derived from more than 20 satellite-pairs per epoch. These samples span a wide range of elevation and azimuth angles, providing a statistically sufficient basis to characterize the receiver noise.

Before conducting the simulations, we first analyze the sensitivity of different constellations to atmospheric errors, based on the analysis framework outlined in previous section. According to Eqs. (16), (17), and (20), the coefficients corresponding to various error sources at each step of the TCAR process can be calculated for each constellation, as summarized in Tables 1 and 3. By

utilizing these coefficients, we can determine the threshold ranges of ionospheric errors that can be considered negligible at each step of TCAR for each constellation, assuming that noise and other errors are absent. This analysis indicates that, provided the residual double-difference ionospheric error remains within these thresholds and noise is ignored, the ambiguities at each step can still be correctly resolved.

These threshold ranges are presented in Table 4. For the EWL ambiguity, both GPS and BDS demonstrate almost the same sensitivity to ionospheric errors, under the given three-frequency. The ambiguity can be reliably resolved provided the residual ionospheric error is below 75.66 TECU. For Galileo, this threshold is slightly lower, at 64.10 TECU. For the WL ambiguity, GPS is the most sensitive to ionospheric errors, with a tolerable limit of no more than 6.10 TECU. In contrast, BDS and Galileo can withstand ionospheric errors up to approximately 8 TECU. For the original frequency ambiguity, all three constellations exhibit similar tolerances, with the maximum allowable ionospheric error limited to around 0.25 TECU.

It is important to note that these threshold ranges are derived under ideal conditions, without considering the presence of noise. Therefore, it is necessary to evaluate the practical significance of these analyses under real-world conditions. The simulated ranges of tropospheric and ionospheric errors are summarized in Table 5. These errors will be introduced into actual GNSS data, which contains real-world measurement noise.

Table 4 Ideal tolerable ionospheric error ranges in TECU for TCAR of GPS, Galileo and BDS (Neglecting Code and Carrier Noise)

Ambiguity type	EWL	WL	NL
GPS	−75.66 to 75.66	−6.10 to 6.10	−0.252 to 0.252
Galileo	−64.10 to 64.10	−8.28 to 8.28	−0.256 to 0.256
BDS	−75.66 to 75.66	−8.59 to 8.59	−0.255 to 0.255

Table 3 Coefficients of different error sources affecting ambiguity resolution for Galileo and BDS

Coefficients		of δT in cycle/m	of E_{TEC} in cycle/TECU	of ϵ_1 in cycle/m	of ϵ_2 in cycle/m	of ϵ_3 in cycle/m	of ϵ_1 in cycle/m	of ϵ_2 in cycle/m	of ϵ_3 in cycle/m
Galileo	EWL	0	0.0078	−0.0796	−0.0796	−0.0796	0	4.27	−4.03
	WL	0	−0.0604	0	0	0	5.26	−21.94	16.68
	NL	0	1.9522	0	0	0	−18.54	5.49	13.05
BDS	EWL	0	0.0066	−0.0682	−0.0682	−0.0682	0	4.23	−4.03
	WL	0	−0.0582	0	0	0	5.21	−24.40	19.19
	NL	0	1.9605	0	0	0	−19.04	6.12	12.95

Table 5 Simulated residual tropospheric and ionospheric errors

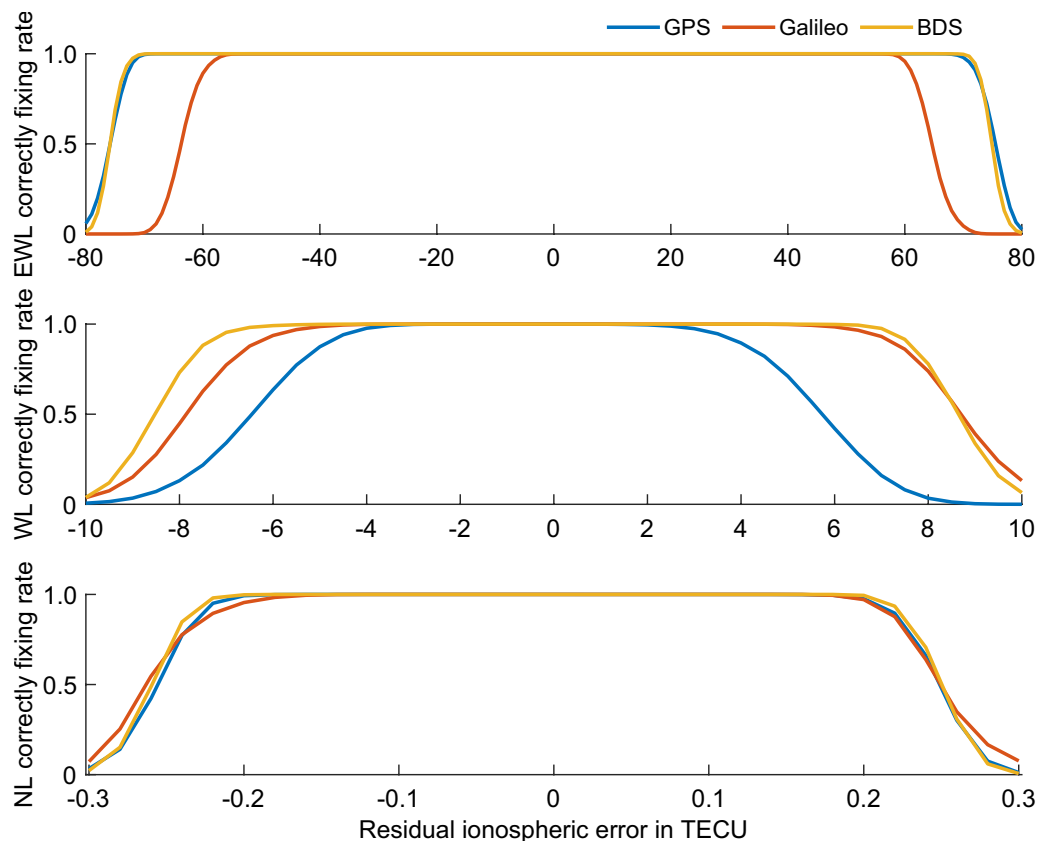
Error source		Range	Interval
Tropospheric error (m)		− 0.5 to 0.5	0.1
Ionospheric error in TECU	EWL	− 80 to 80	1
	WL	− 10 to 10	0.5
	NL	− 0.3 to 0.3	0.02

Figure 2 illustrates the correct fixing rates for EWL, WL, and NL ambiguities under varying simulated ionospheric error conditions. Several insights can be drawn from these results. First, the impact of stochastic noise on the ionospheric error tolerance is evident. Compared to the theoretical value without noise in Table 4, the tolerance of each ambiguity resolution step to ionospheric errors is slightly reduced in the presence of code and carrier phase noises. This indicates that measurement noise reduces the safety margin available for ionospheric errors. However, the reduction is marginal, confirming that the TCAR method maintains strong robustness even under real-world noise levels.

Second, the consistency across constellations validates the universality of the proposed framework. The relative sensitivity of different constellations (GPS, Galileo and BDS) to ionospheric errors aligns with the theoretical derivation. This suggests that the ambiguity resolution performance is dominantly governed by the signal frequency structure and wavelength characteristics, rather than minor variations in receiver specific noise floors.

Third, clear ionospheric error thresholds for different ambiguities are quantified:

- 1) For EWL ambiguities: The correct fixing rate remains at 100% even when residual ionospheric errors reach several tens of TECU. This highlights the robustness of EWL ambiguity as a reliable foundation for the TCAR process.
- 2) For WL ambiguities: The stability threshold tightens to a few TECU. While less robust than EWL, it still falls within the accuracy range of some standard global ionospheric models.
- 3) For NL ambiguities: The ionospheric tolerance is slightly below 0.2 TECU, a value lower than the theoretical limit due to the influence of measurement noise. This indicates that when ionospheric residuals

**Fig. 2** Correct fixing rates of EWL, WL, and NL ambiguities under simulated ionospheric errors

are not fully mitigated, the NL resolution represents the most challenging step in the TCAR process.

When tropospheric error is simulated from -0.5 to 0.5 m, all constellations achieve 100% correct fixing rates for EWL, WL, and NL ambiguities, even in the presence of real-world noise. This result is consistent with previous analyses. This finding is particularly important for RTK applications involving large height differences or long baselines, where tropospheric errors are difficult to be mitigated completely. This is a significant advantage of the GF model over the GB model. Based on these results, we can conclude that residual tropospheric errors have negligible impact on ambiguity resolution at each step of the GF TCAR process.

Field experiments

The GNSS data used in this section were obtained from the Continuously Operating Reference Stations (CORS) network, maintained by the Lands Department of Hong Kong. In cases 1 and 2, the stations HKOH, HKPC, and HKKT form a triangular network, which serves as the basis for interpolating atmospheric delays across this region. HKSC is designated as the rover station, as illustrated in Fig. 3. Although the network coverage is relatively limited, atmospheric errors cannot be neglected due to the active ionosphere of low-latitude regions (Xu & Chen, 2018). All four stations are equipped with the same hardware, i.e., Leica AR25.R4 + LEIT antennas and Leica GR50 receivers.

To validate the findings obtained from the previous analyses and simulations, and to further assess the performance of the GF TCAR model in network RTK applications, GNSS data were collected during two periods: 04:00–05:00 and 12:00–13:00 local time on 10 June 2024.

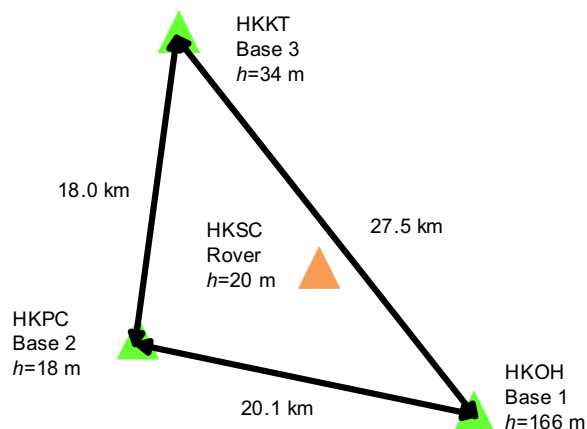


Fig. 3 Configuration of reference and rover stations in the network RTK system

These periods were selected to reflect different levels of ionospheric activity, from quiet (04:00–05:00) to active (12:00–13:00). Although each case consists of only one hour of data, the total number of samples exceeds 36,000, with more than 10 satellite pairs per epoch available on average. Due to the lack of multi-frequency data for BDS, only GPS, Galileo, and the Quasi-Zenith Satellite System (QZSS) are included in this section. The detailed experimental configurations are presented in Table 6. Since the three frequencies used for Galileo differ from those in the previous section, the corresponding error coefficients were recalculated, as shown in Table 7. The excessive length of the EWL for Galileo results in significantly larger carrier phase noise during WL ambiguity resolution compared to GPS and QZSS, suggesting that the correct fixing rate for Galileo WL is expected to be the lowest among three constellations. Atmospheric corrections were generated using the Master-Auxiliary Concept (Park & Kee, 2010), with HKOH designated as the master station.

Analysis of case 1

A brief overview of the procedure for obtaining atmospheric corrections in this network RTK configuration is provided as follows. First, the double-difference ambiguities on the baselines HKOH–HKPC and HKOH–HKKT are resolved. Subsequently, using the known ambiguities and station coordinates, the double-difference slant tropospheric wet delays and ionospheric delays are estimated

Table 6 The details of the experimental setting in cases 1 and 2

Base stations	Antenna: Leica AR25.R4 + LEIT; Receiver: Leica GR50		
Rover station	Antenna: Leica AR25.R4 + LEIT; Receiver: Leica GR50		
Ionospheric error	Linear interpolation		
Tropospheric error	Linear interpolation		
Mask angle	20 degrees		
Local time	Case 1: 4.00–5.00; Case 2: 12.00–13.00		
Sampling rate	1 Hz		
Constellation	QZSS	Galileo	GPS
Carrier frequency	f_1 : 1575.42 MHz f_2 : 1227.60 MHz f_3 : 1176.45 MHz	f_1 : 1575.42 MHz f_2 : 1207.14 MHz f_3 : 1176.45 MHz	f_1 : 1575.42 MHz f_2 : 1227.60 MHz f_3 : 1176.45 MHz
EWL	$\lambda_{(0,1,-1)}$: 5.8610 m	$\lambda_{(0,1,-1)}$: 9.7684 m	$\lambda_{(0,1,-1)}$: 5.8610 m
WL	$\lambda_{(1,-1,0)}$: 0.8619 m	$\lambda_{(1,-1,0)}$: 0.8140 m	$\lambda_{(1,-1,0)}$: 0.8619 m

Table 7 Coefficients of different error sources affecting ambiguity resolution for Galileo

Coefficients		of δT in cycle/m	of E_{TEC} in cycle/TECU	of ε_1 in cycle/m	of ε_2 in cycle/m	of ε_3 in cycle/m	of ϵ_1 in cycle/m	of ϵ_2 in cycle/m	of ϵ_3 in cycle/m
Galileo	EWL	0	0.0041	0.0341	0.0341	0.0341	0	4.03	3.92
	WL	0	0.0883	0	0	0	5.26	52.35	47.09
	NL	0	1.9848	0	0	0	− 16.17	6.81	9.37

based on Eq. (15), without the use of any mapping function. This approach is particularly advantageous for mitigating the inaccuracies that mapping functions may introduce in different directions. The single-epoch slant tropospheric and ionospheric delays for HKOH–HKPC are presented in Fig. 4, as an example. These results demonstrate that, despite the baseline length of 20.1 km, the double-difference slant tropospheric and ionospheric delays remain significant in the Hong Kong, likely due to the complex atmospheric conditions characteristic of low latitudes.

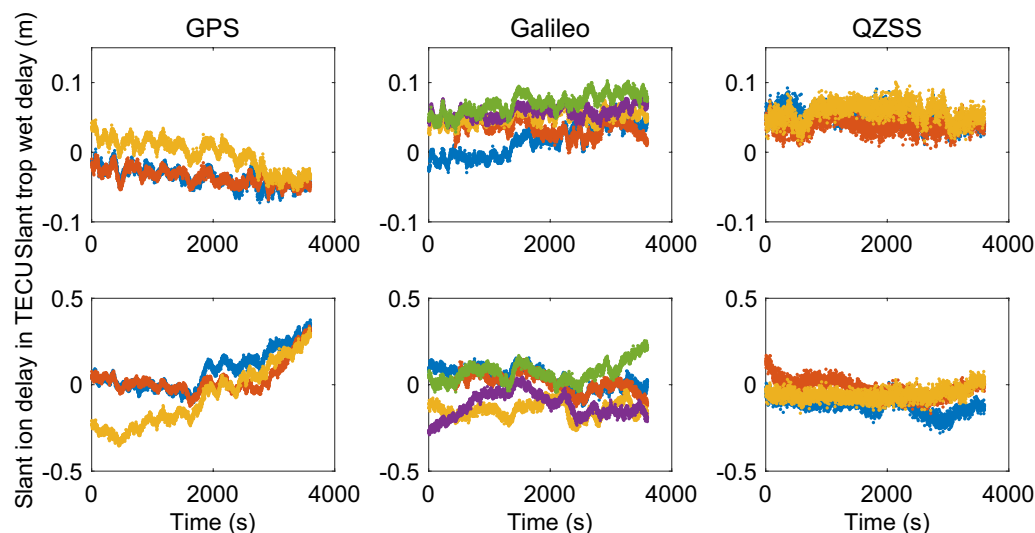
After smoothing, the estimated atmospheric delays are used to interpolate the corrections for the baseline HKOH–HKSC through a linear interpolation. The residual atmospheric delays on the HKOH–HKSC after applying these corrections are shown in Fig. 5. The residual slant tropospheric delays are within 3 cm, and the residual slant ionospheric delays are within 0.1 TECU. According to the previous analysis, the accuracy of these corrections is sufficient to satisfy the requirements for reliable GF TCAR ambiguity resolution.

The number of incorrectly fixed ambiguities for the GF TCAR model is summarized in Table 8. GF TCAR demonstrates excellent performance in Case 1, achieving

nearly 100% correct fixing rates. All EWL ambiguities for the three constellations are correctly resolved by TCAR. For WL ambiguities, the number of incorrect fixes is zero for GPS and one for QZSS, whereas for Galileo, this number increases to 17. This can be attributed to the longer EWL wavelength associated with Galileo in this case. This finding corroborates the earlier analysis, indicating that a longer EWL wavelength adversely affects WL ambiguity resolution. Furthermore, the number of incorrectly fixed ambiguities for the original frequency is identical to that observed for WL, suggesting that WL ambiguity resolution may constitute the most challenging step in TCAR when residual ionospheric error is negligible.

Analysis of case 2

Compared to case 1, the GNSS data were collected at noon, when ionospheric activity is higher. As a result, the ionospheric delays presented in Fig. 6 are larger than those observed in case 1. The maximum ionospheric delay even reached 1 TECU for a 20.1 km baseline. In this case, even after applying corrections, the residual atmospheric delays on baseline HKOH–HKSC remain relatively large (Fig. 7), with the maximum

**Fig. 4** The double-difference slant tropospheric wet and ionospheric delays of GPS, Galileo and QZSS in baseline HKOH–HKPC (4.00–5.00 local time)

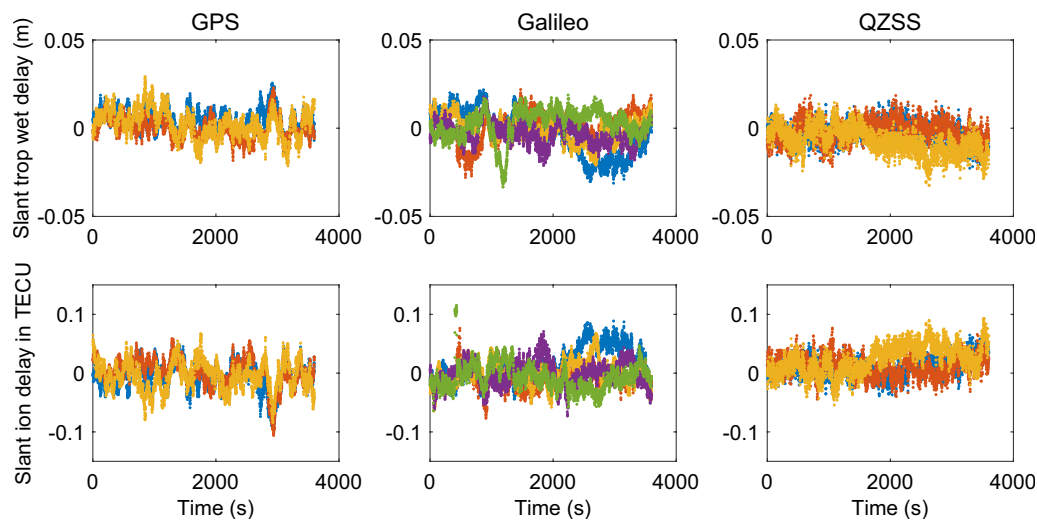


Fig. 5 The double-difference slant tropospheric wet and ionospheric residual delays of GPS, Galileo and QZSS in baseline HKOH-HKSC after corrections (4.00–5.00 local time)

Table 8 The number of incorrectly fixed ambiguities and total ambiguity number for GPS, Galileo and QZSS (4.00–5.00 local time)

Constellation	Ambiguity type	Incorrectly fixed ambiguity number	Total ambiguity number
GPS	EWL	0	10,800
	WL	0	
	NL	0	
Galileo	EWL	0	17,537
	WL	17	
	NL	17	
QZSS	EWL	0	10,800
	WL	1	
	NL	1	

tropospheric residual reaching approximately 0.1 m and the maximum ionospheric residual about 0.2 TECU. Atmospheric gradients in different directions can reduce the accuracy of the interpolated atmospheric delays. Additionally, the residuals for GPS and Galileo are larger than those for QZSS, likely because QZSS satellites generally maintain higher and more stable elevation angles.

The detailed ambiguity fixing results in case 2 are summarized in Table 9. As shown, all EWL ambiguities are correctly fixed across all constellations, achieving a 100% correct fixing rate. For WL ambiguities, the number of incorrectly fixed cases is 193 for Galileo, while 0 for GPS and 2 for QZSS. This discrepancy is primarily attributed to the excessively long wavelength of the

EWL combination used for Galileo in this test. With more appropriate frequency combinations, improved performance for Galileo can be anticipated.

The most noteworthy finding is the fixing performance of the original frequency ambiguities. All three constellations achieve nearly 100% correct fixing rates for the original frequency, provided that the EWL and WL ambiguities are correctly resolved. This result further demonstrates that WL may be the most difficult step, if residual ionospheric error is negligible.

Analysis of case 3

Case 3 utilizes Virtual Reference Station (VRS) services provided by the Lands Department of Hong Kong (https://www.geodetic.gov.hk/sc/satref/Net_RTK.htm) on 31 March 2026. To ensure a rigorous evaluation, an independent receiver was designated as the rover station, which was not involved in the generation of the VRS data. A continuous 24-h dataset encompassing various levels of ionospheric activity was employed to validate the proposed analytical framework. The detailed configurations for this experiment are summarized in Table 10.

The ambiguity resolution results are presented in Table 11 and Fig. 8. Overall, the majority of ambiguities at each step were correctly fixed, demonstrating the effectiveness of the VRS service. Under a 20-degree mask angle and 35 dB-Hz C/N_0 threshold, 4, 176, and 877 incorrect fixes were identified for the EWL, WL, and NL ambiguities, respectively. When the mask angle was lowered to 10 degrees with a 30 dB-Hz C/N_0 threshold, these values rose to 4, 401, and 1,194. Specifically, the number of incorrect WL fixes increased by 277, reflecting higher noise levels at lower elevation angles.

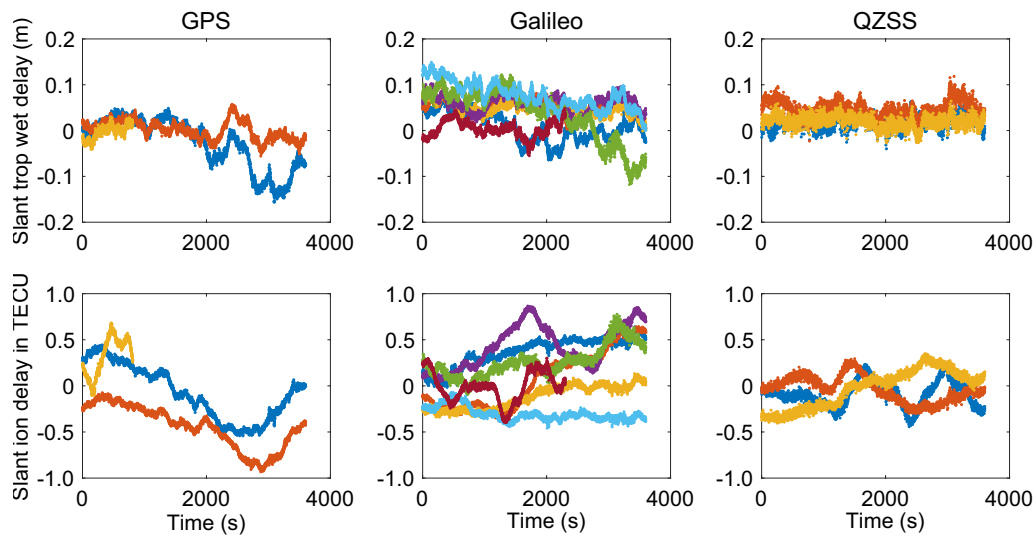


Fig. 6 The double-difference slant tropospheric wet and ionospheric delays of GPS, Galileo and QZSS in baseline HKOH-HKPC (12.00–13.00 local time)

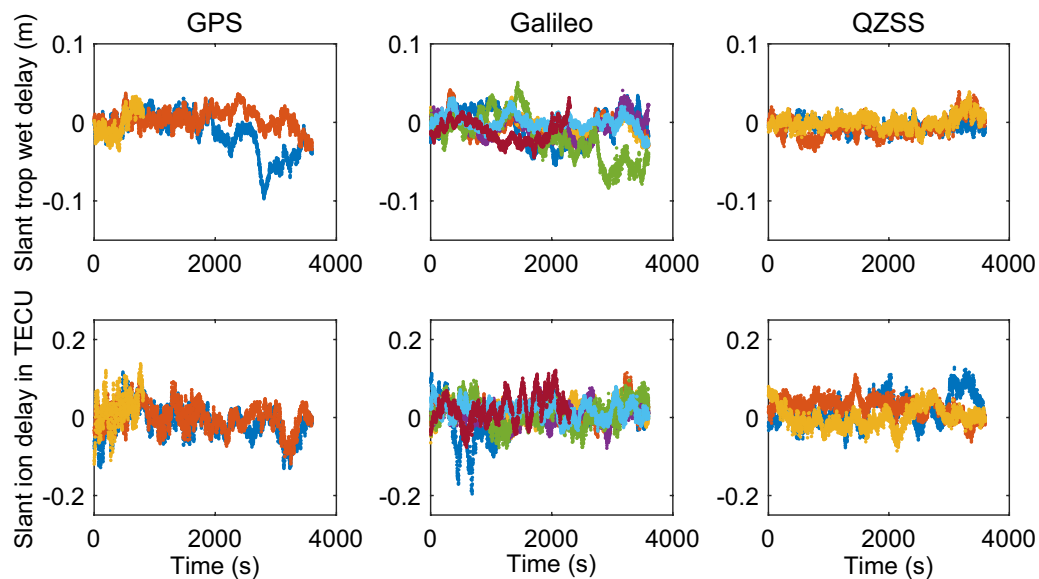


Fig. 7 The double differenced slant tropospheric wet and ionospheric residual delays of GPS, Galileo and QZSS in baseline HKOH-HKSC after corrections (12.00–13.00 local time)

Regarding the N_2 ambiguity, although the total number of incorrect fixes increased by 317, only 40 of these occurred specifically during the NL resolution process (317 minus the 277 inherited from WL). This suggests that the ionospheric corrections remain sufficiently accurate even at a 10-degree mask angle. The range errors caused by incorrect fixes are shown in Fig. 8. From 0:00 to 12:00 UTC, nearly all incorrect fixes occurred during the WL step, indicating that residual ionospheric errors from the VRS service remained within 0.2 TECU. In

contrast, between 12:00 and 18:00 UTC (corresponding to 20:00–02:00 local time in Hong Kong), the AR success rate notably decreased due to ionospheric scintillation (Yang et al., 2018). As year 2026 approaches the time of solar maximum of Cycle 25, ionospheric scintillation in low-latitude regions like Hong Kong is expected to reach a decadal peak (Chowdhury et al., 2021). During these scintillation events, even four EWL ambiguities were incorrectly resolved, highlighting the severe

Table 9 The number of incorrectly fixed ambiguities and total ambiguity number for GPS, Galileo and QZSS (12.00–13.00 local time)

Constellation	Ambiguity type	Incorrectly fixed ambiguity number	Total ambiguity number
GPS	EWL	0	8011
	WL	0	
	NL	3	
Galileo	EWL	0	23,903
	WL	193	
	NL	194	
QZSS	EWL	0	10,800
	WL	2	
	NL	2	

Table 10 The details of the experimental setting in case 3

Base stations	VRS from Lands Department, Hong Kong		
Rover station	Receiver: Septentrio PolaRx5e		
Sampling duration	24 h		
Sampling interval	30 s		
Constellation	BDS (II)	Galileo	GPS
Carrier frequency	f_1 : 1561.098 MHz f_2 : 1268.52 MHz f_3 : 1207.14 MHz	f_1 : 1575.42 MHz f_2 : 1207.14 MHz f_3 : 1176.45 MHz	f_1 : 1575.42 MHz f_2 : 1227.60 MHz f_3 : 1176.45 MHz
EWL	$\lambda_{(0,1,-1)}$: 4.8842 m	$\lambda_{(0,1,-1)}$: 9.7684 m	$\lambda_{(0,1,-1)}$: 5.8610 m
WL	$\lambda_{(1,-1,0)}$: 1.0247 m	$\lambda_{(1,-1,0)}$: 0.8140 m	$\lambda_{(1,-1,0)}$: 0.8619 m

Table 11 The number of incorrectly fixed ambiguities and total ambiguity number for three constellations under different mask angles

Ambiguity type	Incorrectly fixed ambiguity number with 10° mask angle	Incorrectly fixed ambiguity number with 20° mask angle	Total ambiguity number
EWL	4	4	41,563 (10° mask angle)
WL	401	176	36,267 (20° mask angle)
NL	1194	877	

adverse impact of intense ionospheric disturbances on AR performance.

To mitigate the impact of incorrectly fixed ambiguities on positioning accuracy, a Chi-square test was employed

to identify and exclude affected epochs (Cheng et al., 2024). The resulting positioning performance is detailed in Fig. 9 and Table 12. As illustrated in Fig. 9, high-precision positioning could not always be maintained in the period between 12:00 and 18:00 UTC due to the severe influence of ionospheric scintillation.

To better evaluate the positioning performance, the 24-h dataset was divided into three 8-h segments based on ionospheric activity levels, including the quiet period (0–2 h and 18–24 h), the active period (2–10 h), and the scintillation period (10–18 h). As summarized in Table 12, positioning availability generally decreases slightly when a 10-degree mask angle is applied. This is primarily due to the increased occurrence of incorrectly fixed ambiguities, which are excluded from the positioning process by the Chi-square test.

Regarding positioning accuracy, the performance varies across different ionospheric conditions. During the quiet period, the 3D Root Mean Square Error (RMSE) at a 10-degree mask angle (1.62 cm) is lower than that at 20-degree (1.76 cm), mainly because the inclusion of more low-elevation satellites enhances the satellite geometry. In contrast, during the active period, the 10-degree mask angle yields a 3D RMSE of 2.49 cm, slightly higher than the 2.31 cm for the 20-degree mask angle. This discrepancy arises from larger ionospheric gradients, which complicate the accurate interpolation of ionospheric delays, particularly for low-elevation satellites; although these residual errors may not impact ambiguity resolution, they still propagate into the final positioning solution. During the scintillation period, the 3D RMSE for the 10-degree mask angle (5.16 cm) is again lower than that for the 20-degree mask angle (5.55 cm). This improvement is attributed to both the strengthened geometry and the rejection of epochs with significant positioning errors by the Chi-square test.

These results demonstrate that the VRS service provided by the Hong Kong Lands Department exhibits performance comparable to that of a short-baseline setup. Under non-scintillation periods, the residual atmospheric errors satisfy the accuracy requirements established in our previous analysis.

Discussion of experiments

The sequence of experiments in cases 1 to 3 provides a robust validation of the proposed analysis framework, transitioning from a basic triangular network to a commercial VRS application.

Cases 1 and 2 utilized a simplified triangular network to test the impact of residual ionospheric errors under different conditions. The estimated residual ionospheric errors after applying the corrections of the both cases were almost entirely within the analyzed threshold (0.2

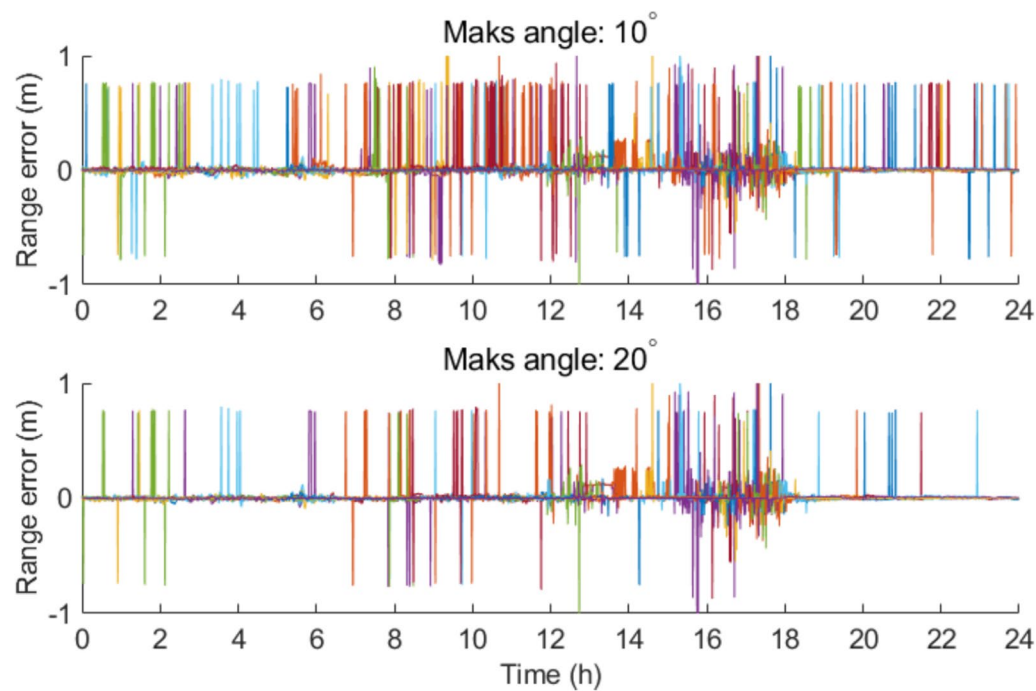


Fig. 8 Carrier phase range error after TCAR using VRS in case 3 (UTC)

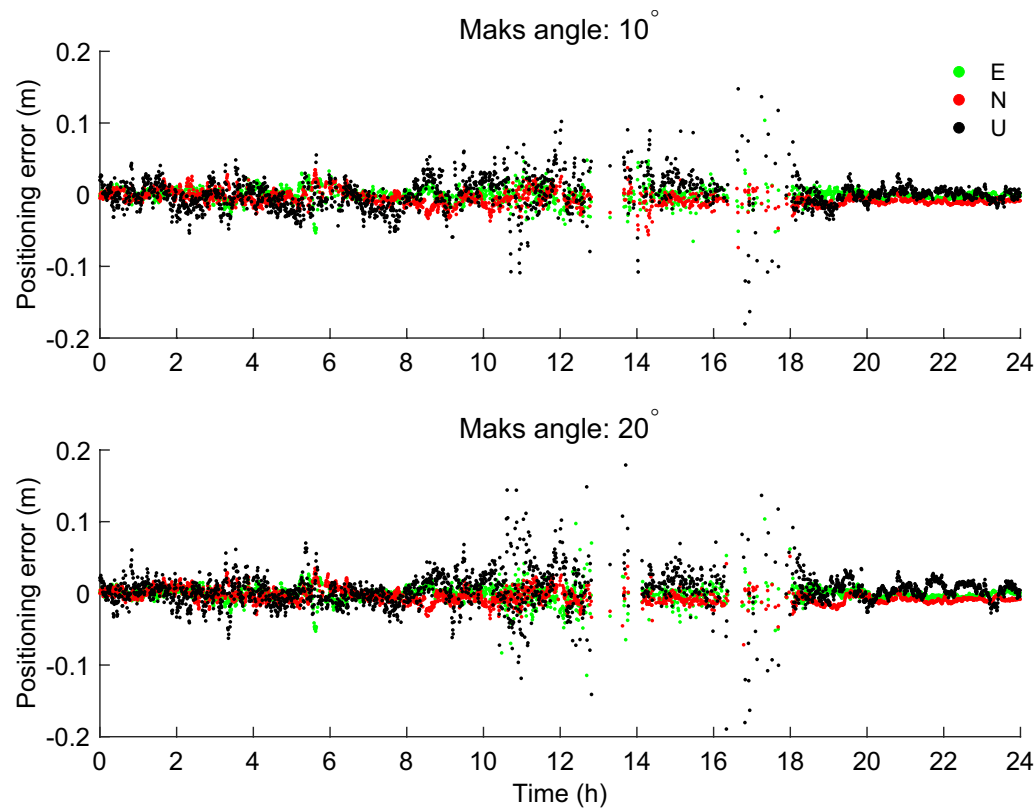


Fig. 9 Positioning errors in east, north, and up components of TCAR using VRS in case 3 (UTC)

Table 12 Positioning epoch number and 3D RMSE of TCAR using VRS in case 3

Time period	Ionospheric quiet period (0-2h and 18-24h)		Ionospheric active period (2-10h)		Ionospheric scintillation period (10-18h)	
Mask angle	Number of epochs	3D RMSE (cm)	Number of epochs	3D RMSE (cm)	Number of epochs	3D RMSE (cm)
10°	853 (0.889)	1.62	790 (0.823)	2.49	424 (0.442)	5.16
20°	929 (0.968)	1.76	898 (0.935)	2.31	495 (0.516)	5.55

TECU). The correct fixing rates for EWL, WL, and NL were all near 100%, demonstrating the effectiveness of the proposed framework and the validity of the derived ionospheric thresholds.

Applying a similar analysis to case 3, which involved the Hong Kong Lands Department's professional-grade VRS service, demonstrates the practical utility of proposed analysis framework as an evaluation tool for commercial network RTK performance. The results show that for most of time in the 24-h period, the commercial VRS service easily satisfies the 0.2 TECU threshold required for centimeter-level positioning. Even when the ionospheric delay reached its peak around 06:00 UTC (14:00 local time), the VRS performance remained stable. This confirms that the Hong Kong VRS service is reliable under standard atmospheric conditions, where the ionosphere activity is elevated but stable.

Another key finding from case 3 is the theoretical explanation for the performance degradation of network RTK services during ionospheric scintillation. Network RTK and VRS rely on the assumption of spatial correlation in the ionosphere to interpolate corrections from reference stations to the rover. However, ionospheric scintillation creates rapid and small-scale stochastic fluctuations that destroy this correlation. The proposed framework can analyse this failure. During the 12:00–18:00 UTC period, scintillation-induced ionospheric residuals led to a lower NL correctly fixing rate. However, the correct fixing rates for EWL and WL remained relatively high, indicating that the ionospheric residuals were likely between 0.2 and several TECU. This also suggests that the ionospheric residual, rather than enlarged noise, is the primary obstacle during ionospheric scintillation. As illustrated in Fig. 1, if noise were the dominant factor, the correct fixing rates for WL and NL would be close.

Besides ionospheric scintillation, other events, such as during geomagnetic disturbances in equatorial and polar regions, can produce ionospheric conditions characterized by small-scale irregularities and steep gradients that compromise its spatial correlation. This instability, rather than the absolute magnitude of the delay, significantly degrades the interpolation accuracy of ionospheric

corrections, making NL ambiguity resolution challenging. While the availability of centimeter-level NL resolution may be difficult to maintain during such periods, GF TCAR may remain robust for EWL and WL combinations due to their lower sensitivity to ionospheric residuals. This allows for a reliable fallback to decimeter to meter-level positioning even under severe ionospheric conditions.

Conclusion

This study proposes an enhanced analysis framework and presents a comprehensive analysis of the impact of residual atmospheric errors and measurement noise on the performance of the GF TCAR model in network RTK, with a particular focus on ambiguity resolution. Several key findings emerge from theoretical analysis, simulation and field experiment results.

First, the GF TCAR framework demonstrates less sensitivity to residual non-dispersive errors. The tropospheric delays have negligible effect on ambiguity resolution at each stage of cascading process. This characteristic is particularly advantageous for RTK applications in environments with significant height differences or long baselines, where tropospheric delay cannot be completely mitigated by the double-difference.

Second, residual ionospheric errors are identified as the primary constraint on ambiguity resolution. We estimated tolerance thresholds for different ambiguities. EWL and WL ambiguities exhibit strong robustness, maintaining fixing rates of nearly 100% even when residual errors reach tens of TECU and several TECU, respectively. In contrast, reliable resolution of original frequency (NL) ambiguities requires a higher accuracy of within ± 0.2 TECU. This value serves as a critical benchmark for evaluating the quality of ionospheric corrections or ionospheric models.

Third, the sensitivity to measurement noise varies significantly across different combinations. While the EWL combination demonstrates strong robustness to noise, the WL combination is more sensitive to it, particularly as the EWL wavelength increases. When Gaussian noise with a standard deviation of 0.03 m is added to the three carrier phase measurements, EWL ambiguities can still

be correctly fixed with a rate of 100%. In contrast, WL ambiguities can be nearly 100% correctly fixed only when the standard deviation of the additional noise does not exceed 0.005 m. When both EWL and WL ambiguities are correctly fixed, the probability of correctly fixing the original frequency ambiguity is also very high. Consequently, minimizing carrier phase noise is a prerequisite for reliable WL ambiguity resolution.

Overall, these findings validate the proposed analytical framework and provide quantitative guidance for optimizing multi-frequency GNSS ambiguity resolution, especially in low-latitude regions characterized by active ionospheric conditions. Furthermore, the results offer practical benchmarks for assessing ionospheric correction quality and system robustness under varying atmospheric states.

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Author contributions

Q. C. proposed the method, processed the data and wrote the main manuscript. All authors reviewed the manuscript.

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Availability of data and material

Dataset is available on the web site (<https://www.geodetic.gov.hk/en/rinex/download.aspx>).

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no conflict of interest.

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